

Orwa Jabali

AI Engineer | M.Sc. Artificial Intelligence | Multi-Agent Systems · LLMs · RAG Pipelines

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PROFESSIONAL SUMMARY

AI Engineer with an M.Sc. in Artificial Intelligence, specializing in Multi-Agent Systems, Large Language Models (LLMs), and Retrieval-Augmented Generation (RAG). Experienced in designing and deploying end-to-end AI pipelines — from vector-based document retrieval and LLM orchestration to real-time data streaming. Combines strong research foundations with practical software engineering skills. Eager to contribute to international AI teams building intelligent, scalable systems.

TECHNICAL SKILLS

Domain	Technologies
Languages	Python, Scala, C#, C++
AI / LLM	LangChain, LangGraph, LangSmith, HuggingFace Transformers, RAG, Multi-Agent Systems, NLP
ML / Deep Learning	PyTorch, Scikit-learn, Computer Vision, CNNs, Transfer Learning
Vector Databases	Pinecone, ChromaDB
Big Data	Apache Spark, Apache Kafka
Web / APIs	FastAPI, Flask, Streamlit
DevOps / Tools	Docker, Git, GitHub, Jupyter Notebook
Databases	SQL basics, NoSQL MongoDB, Pinecone & ChromaDB (vector store)

EDUCATION

M.Sc. in Artificial Intelligence

An-Najah National University — Nablus, Palestine | 2023 – 2025

- Thesis: *Grounding LLMs in Pedagogical Context: A Multi-Agent Framework for Cross-Domain Educational Assessment*
- Grade: 3.05 / 4

B.Sc. in Building Engineering

An-Najah National University — Nablus, Palestine | 2007 – 2012

- Grade: 78.2 / 100

KEY PROJECTS

Multi-Agent RAG Chatbot | 2025

github.com/orwajabali/multi-agent-rag-chatbot

- Designed and built a production-grade multi-agent conversational AI system from the ground up.
 - Implemented a RAG pipeline with a FAISS vector store for efficient semantic document retrieval.
 - Orchestrated agent workflows using LangGraph with LLaMA 3.1 (70B) as the backbone LLM.
 - Enabled multi-turn conversation memory and bilingual (Arabic & English) document Q&A.
 - Deployed with a Docker-containerized FastAPI backend; full CI/CD pipeline via GitHub.
 - **Stack:** Python · LangGraph · LangChain · LangSmith · LLaMA 3.1 · FAISS · FastAPI · Docker
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SafeRoute Stream — Real-Time Traffic Safety Pipeline | 2024

github.com/orwajabali/saferoute-stream

- Engineered a real-time data streaming pipeline for route safety scoring using event-driven architecture.
 - Applied Apache Kafka for message brokering and Apache Spark for distributed stream processing.
 - **Stack:** Python · Apache Kafka · Apache Spark
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Scientific Paper Title Recovery — NLP | 2024

github.com/orwajabali/scientific-paper-title-recovery

- Developed an NLP model to recover and predict masked titles from scientific paper abstracts.
 - Applied transformer-based sequence-to-sequence modelling and fine-tuning techniques.
 - **Stack:** Python · PyTorch · HuggingFace Transformers
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Cat Breed Classification — Computer Vision | 2023

github.com/orwajabali/cat-breed-classification

- Comparative study: custom CNN achieved 75.98% accuracy vs YOLOv8 + LightGBM reaching 81.81%.
 - Benchmarked multiple architectures using transfer learning and feature engineering pipelines.
 - **Stack:** Python · PyTorch · YOLOv8 · LightGBM · OpenCV
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Additional Projects — 2023

Project	Description	Stack	Link
Text Classification — 91 Classes	Multi-class classifier across 91 document categories using TF-IDF + SVM	Python · Scikit-learn · NLTK	↗
UK Road Safety Mining	Data mining on 8.6M accident records — EDA, feature engineering, association rules	Python · Pandas · Matplotlib	↗
Rain Prediction Benchmark	Benchmarked 9 classifiers on 25K records; ROC-AUC up to 1.00 with ensemble methods	Scikit-learn · XGBoost · LightGBM	↗
House Price Prediction	Linear, Ridge & Lasso regression implemented from scratch	Python · NumPy	↗
Temperature Forecasting	10-year Melbourne temperature forecast using Facebook Prophet	Python · Prophet · Pandas	↗
Document Clustering	K-Means clustering on 4,601 text documents for unsupervised topic grouping	Python · Scikit-learn	↗
Customer Behavior Mining	Purchase pattern mining using Apriori & FP-Growth for market basket analysis	Python · mlxtend	↗
Developer Survey EDA	Exploratory analysis on 64K+ Stack Overflow developer survey responses	Python · Seaborn · Plotly	↗
Clustering Geometry Study	Benchmarking K-Means, DBSCAN, and GMM across geometric datasets	Python · Scikit-learn	↗
Stereo Disparity Map	Depth estimation comparison: StereoBM vs StereoSGBM algorithms	Python · OpenCV · NumPy	↗

ANN Multiclass Classifier — Real-Time Visualization | 2022

github.com/orwajabali/ann-multiclass-classifier

- Implemented a fully custom Artificial Neural Network with real-time decision boundary visualization.
- Designed an interactive GUI showing live weight updates and class separation during training.
- **Stack:** C# · WinForms

TSP Solver — Simulated Annealing | 2022

github.com/orwajabali/tsp-simulated-annealing

- Solved the NP-hard Traveling Salesman Problem using Simulated Annealing metaheuristic.
- Built an interactive GUI with animated route optimization and temperature decay visualization.
- **Stack:** C# · WinForms

PROFESSIONAL EXPERIENCE

Founder & CEO

Private Contracting Company — Nablus, Palestine | Aug 2013 – May 2021 (*Self-employed*)

- Founded and managed a private company, overseeing day-to-day operations, project planning, budgeting, and client relations across multiple concurrent projects.

- Led cross-functional teams, coordinated with vendors and stakeholders, and drove business development initiatives.
 - Developed strong skills in strategic decision-making, resource management, and operational leadership.
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VOLUNTEER EXPERIENCE

AR/VR Demos & Prototypes

An-Najah Innovation Park — Nablus, Palestine | 6 Months

- Volunteered at the university's innovation hub to design and demonstrate AR/VR prototypes for academic and public audiences.
 - Contributed to hands-on technology showcases promoting emerging tech within the local academic community.
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LANGUAGES

- **Arabic** — Native
 - **English** — Good
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References available upon request.